

ZARI.ai — EfficientNetV2-B2 Model Evaluation Report

Complete Training Results, Evaluation Metrics & Diagnostic Visualizations • Generated: 2026-08-20 12:51 • All data sourced directly from training artifacts — no fabricated values.

1. Final Test Set Evaluation Metrics

These metrics are computed on the held-out **test set (14,260 samples)** using the best checkpoint from epoch 5 (selected by highest validation Macro-F1).

Metric	Value	Description
Test Accuracy	81.80%	Proportion of correct predictions across all 14,260 test images
Test Macro-F1	0.6793	Unweighted mean F1 across all 150 classes (sensitive to rare classes)
Test Weighted-F1	0.7754	Sample-weighted mean F1 (reflects overall prediction quality)
Test Macro-Precision	0.6691	Unweighted mean precision across all classes
Test Macro-Recall	0.7156	Unweighted mean recall across all classes
Test Top-5 Accuracy	99.71%	% of times correct class is in top-5 predictions
Test Loss	0.5688	CrossEntropy loss on test set (unweighted)
Best Epoch	5	Epoch with highest validation Macro-F1 (early stopped at epoch 12)
Test Samples	14,260	Total images in the test split
Num Classes	150	Total crop-disease target classes

2. Training Configuration & Hyperparameters

Parameter	Value	Parameter	Value
Model	tf_efficientnetv2_b2	Pretrained	True
Input Size	260×260	Batch Size	64
Learning Rate	0.001	Weight Decay	0.0001
Label Smoothing	0.1	Gradient Clip	1.0
Optimizer	AdamW	Scheduler	CosineWarmRestart T0=5
Max Epochs	30	Early Stop Patience	7
Train/Val/Test	80% / 10% / 10%	Seed	42
Loss Function	CrossEntropy + Class Weights	Mixed Precision	AMP (float16)
Split Sizes	114,076 / 14,260 / 14,260	Num Workers	8

3. Per-Epoch Training Log (Actual Values)

Every row below is the actual recorded value from training. Bold rows indicate new best checkpoints.

Epoch	Train Loss	Train Acc	Train F1	Val Loss	Val Acc	Val F1	LR	Time (s)
1 ★	2.1556	66.15%	0.5627	1.8075	72.32%	0.6017	0.001000	123.8
2 ★	1.8173	73.97%	0.6419	1.7067	74.61%	0.6170	0.000905	121.3
3 ★	1.7027	77.23%	0.6703	1.6249	80.45%	0.6600	0.000655	122.1
4 ★	1.6201	80.31%	0.6921	1.5773	81.39%	0.6698	0.000346	121.4
5 ★	1.5669	82.57%	0.7051	1.5552	82.04%	0.6834	0.000096	123.0
6	1.7381	75.95%	0.6642	1.6560	79.10%	0.6432	0.001000	122.5
7	1.6943	77.34%	0.6736	1.6449	77.42%	0.6478	0.000976	120.9
8	1.6618	78.29%	0.6855	1.6192	78.44%	0.6555	0.000905	122.1
9	1.6286	79.44%	0.6963	1.6052	78.20%	0.6522	0.000794	122.3
10	1.6011	80.43%	0.7066	1.5861	78.98%	0.6574	0.000655	119.5
11	1.5697	81.43%	0.7117	1.5705	81.76%	0.6765	0.000500	121.3
12	1.5438	82.61%	0.7213	1.5510	81.78%	0.6807	0.000346	121.9

★ = New best checkpoint saved

4. Training Curves (Loss, Accuracy, F1, Learning Rate)

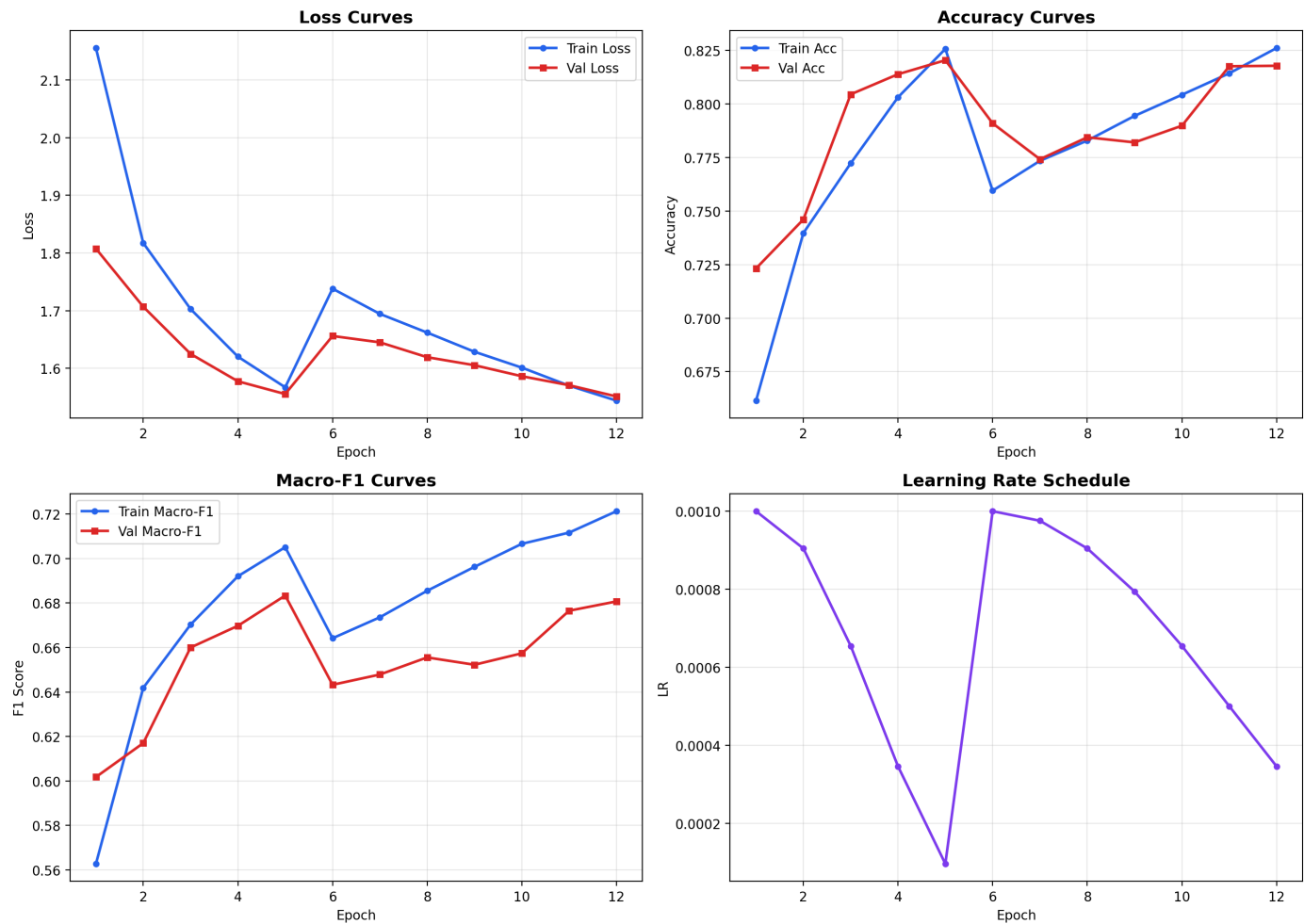


Figure 1: Per-epoch training and validation curves. The cosine LR restart at epoch 6 is visible. Best checkpoint at epoch 5.

5. Per-Class F1 Score Analysis

Performance Tier	F1 Range	# Classes	% of 150
Perfect	F1 = 1.00	23	15.3%
Excellent	$0.90 \leq F1 < 1.00$	36	24.0%
Good	$0.75 \leq F1 < 0.90$	40	26.7%
Moderate	$0.00 < F1 < 0.75$	29	19.3%
Zero / Failed	F1 = 0.00	24	16.0%

Per-Class F1 Distribution (Macro-F1 = 0.6793)

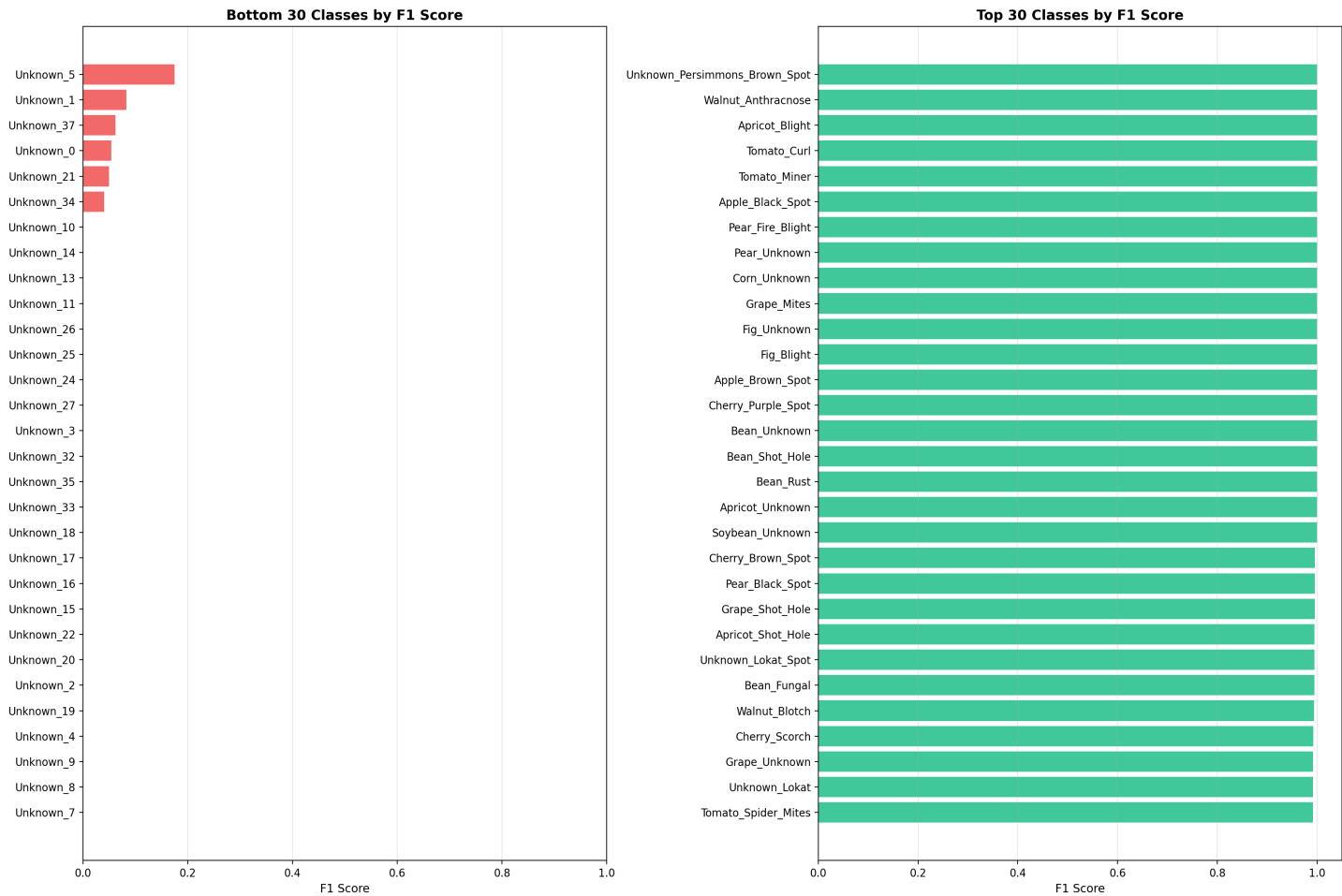


Figure 2: Bottom 30 and Top 30 classes by F1 score. All zero-F1 classes are Unknown_* numeric labels from the NWRD dataset.

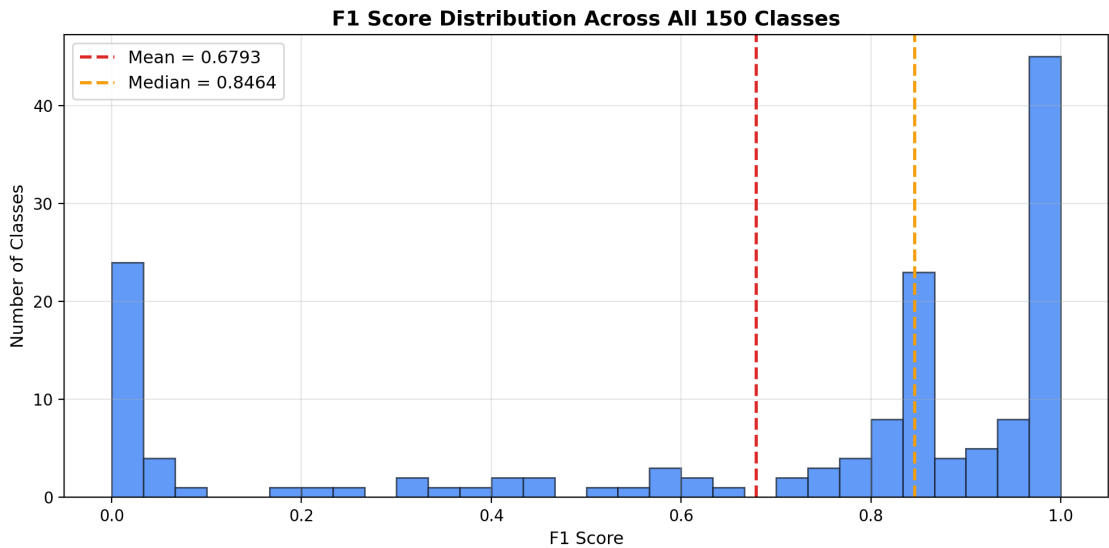


Figure 3: Histogram of F1 scores across all 150 classes showing bimodal distribution.

6. Top 20 Best & Worst Performing Classes (Actual)

Top 20 Best Classes by F1 Score

Class Name	Precision	Recall	F1	Support
Apple_Black_Spot	1.00	1.00	1.00	57
Apple_Brown_Spot	1.00	1.00	1.00	132
Apricot_Blight	1.00	1.00	1.00	40
Apricot_Shot_Hole	0.99	1.00	1.00	104
Apricot_Unknown	1.00	1.00	1.00	81
Bean_Rust	1.00	1.00	1.00	62
Bean_Shot_Hole	1.00	1.00	1.00	49
Bean_Unknown	1.00	1.00	1.00	113
Cherry_Brown_Spot	0.99	1.00	1.00	121
Cherry_Purple_Spot	1.00	1.00	1.00	122
Corn_Unknown	1.00	1.00	1.00	71
Fig_Blight	1.00	1.00	1.00	96
Fig_Unknown	1.00	1.00	1.00	75
Grape_Mites	1.00	1.00	1.00	63
Grape_Shot_Hole	0.99	1.00	1.00	106
Pear_Black_Spot	0.99	1.00	1.00	113
Pear_Fire_Blight	1.00	1.00	1.00	48
Pear_Unknown	1.00	1.00	1.00	96
Soybean_Unknown	1.00	1.00	1.00	7
Tomato_Curl	1.00	1.00	1.00	189

Bottom 20 Worst Classes by F1 Score

Class Name	Precision	Recall	F1	Support
Unknown_15	0.00	0.00	0.00	201
Unknown_16	0.00	0.00	0.00	83
Unknown_17	0.00	0.00	0.00	10
Unknown_18	0.00	0.00	0.00	29
Unknown_19	0.00	0.00	0.00	54
Unknown_2	0.00	0.00	0.00	10
Unknown_20	0.00	0.00	0.00	37
Unknown_22	0.00	0.00	0.00	4
Unknown_24	0.00	0.00	0.00	178
Unknown_25	0.00	0.00	0.00	65
Unknown_26	0.00	0.00	0.00	38
Unknown_27	0.00	0.00	0.00	18
Unknown_3	0.00	0.00	0.00	55
Unknown_32	0.00	0.00	0.00	58
Unknown_33	0.00	0.00	0.00	64
Unknown_35	0.00	0.00	0.00	200
Unknown_4	0.00	0.00	0.00	47
Unknown_7	0.00	0.00	0.00	19
Unknown_8	0.00	0.00	0.00	43
Unknown_9	0.00	0.00	0.00	35

7. Confusion Matrix — Top 25 Most Confused Classes

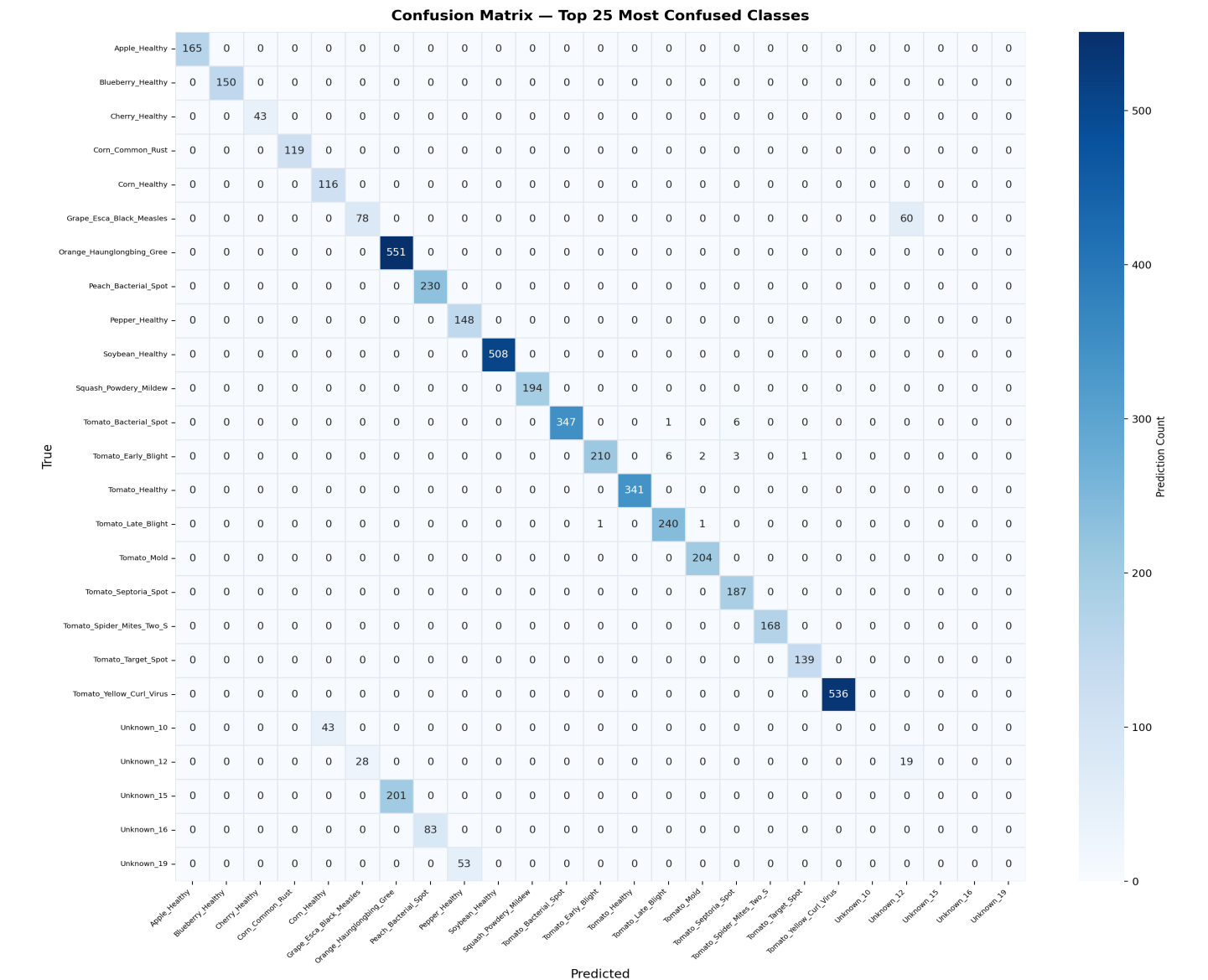


Figure 4: Confusion matrix for the 25 most confused class pairs. Unknown_* classes are systematically misclassified into visually similar named classes.

8. Confidence Score Distribution & Calibration

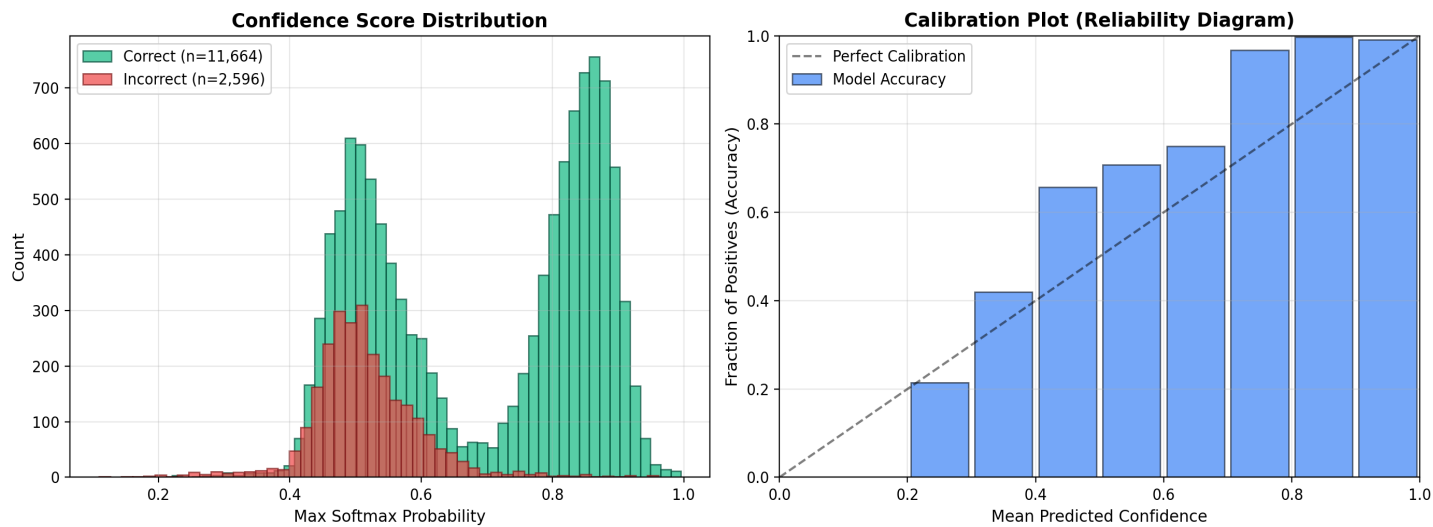


Figure 5: Left — Confidence histogram for correct (green, n=11,664) vs incorrect (red, n=2,596) predictions. Right — Reliability diagram showing the model is slightly overconfident in the 0.5-0.8 range.

9. Full sklearn Classification Report (150 Classes)

Complete per-class precision, recall, F1-score, and support from sklearn.metrics.classification_report.

Class	P	R	F1	N	Class	P	R	F1	N
Apple_Black_Rot	0.76	0.95	0.84	62	Tomato_Late_Blight	0.85	0.70	0.77	343
Apple_Black_Spot	1.00	1.00	1.00	57	Tomato_Minor	1.00	1.00	1.00	190
Apple_Brown_Spot	1.00	1.00	1.00	132	Tomato_Mold	0.98	0.69	0.81	295
Apple_Cedar_Rust	0.72	1.00	0.84	28	Tomato_Mosaic_Virus	0.64	0.17	0.26	42
Apple_Healthy	0.75	1.00	0.86	165	Tomato_Septoria	0.99	0.99	0.99	160
Apple_Rust	0.75	0.67	0.71	9	Tomato_Septoria_Spot	0.73	0.97	0.83	192
Apple_Scab	0.72	0.74	0.73	73	Tomato_Spider_Mites	1.00	0.98	0.99	63
Apple_Unknown	0.99	0.97	0.98	112	Tomato_Spider_Mites_Two_Spotted_Spider_Mite	0.72	1.00	0.84	168
Apricot_Blight	1.00	1.00	1.00	40	Tomato_Target_Spot	0.75	0.99	0.85	140
Apricot_Shot_Hole	0.99	1.00	1.00	104	Tomato_Unknown	0.50	0.43	0.46	7
Apricot_Unknown	1.00	1.00	1.00	81	Tomato_Verticillium_Wilt	0.98	1.00	0.99	52
Bean_Fungal	0.99	1.00	0.99	97	Tomato_Yellow_Curl_Virus	0.73	1.00	0.84	536
Bean_Rust	1.00	1.00	1.00	62	Tomato_Yellow_Virus	0.42	0.71	0.53	7
Bean_Shot_Hole	1.00	1.00	1.00	49	Unknown_0	0.07	0.05	0.05	22
Bean_Unknown	1.00	1.00	1.00	113	Unknown_1	0.25	0.05	0.08	20
Blueberry_Healthy	0.76	1.00	0.86	150	Unknown_10	0.00	0.00	0.00	43
Blueberry_Unknown	0.91	0.91	0.91	11	Unknown_11	0.00	0.00	0.00	40
Cherry_Brown_Spot	0.99	1.00	1.00	121	Unknown_12	0.24	0.40	0.30	47
Cherry_Healthy	0.74	0.50	0.60	86	Unknown_13	0.00	0.00	0.00	38
Cherry_Powdery_Mildew	0.72	0.90	0.80	105	Unknown_14	0.00	0.00	0.00	15
Cherry_Purple_Spot	1.00	1.00	1.00	122	Unknown_15	0.00	0.00	0.00	201
Cherry_Scorch	0.99	0.99	0.99	138	Unknown_16	0.00	0.00	0.00	83
Cherry_Shot_Hole	0.96	1.00	0.98	55	Unknown_17	0.00	0.00	0.00	10
Cherry_Unknown	1.00	0.91	0.95	68	Unknown_18	0.00	0.00	0.00	29
Corn_Blight	0.76	0.84	0.80	19	Unknown_19	0.00	0.00	0.00	54
Corn_Cercospora_Spot_Gray_Spot	0.70	0.92	0.79	52	Unknown_2	0.00	0.00	0.00	10
Corn_Common_Rust	0.73	1.00	0.85	119	Unknown_20	0.00	0.00	0.00	37
Corn_Fungal	0.98	1.00	0.99	41	Unknown_21	0.33	0.03	0.05	37
Corn_Gray_Spot	0.99	0.93	0.96	73	Unknown_22	0.00	0.00	0.00	4
Corn_Healthy	0.73	1.00	0.84	116	Unknown_23	0.42	0.48	0.45	23
Corn_Holcus_Spot	0.98	1.00	0.99	54	Unknown_24	0.00	0.00	0.00	178
Corn_Northern_Blight	0.73	0.99	0.84	99	Unknown_25	0.00	0.00	0.00	65
Corn_Rust	1.00	0.83	0.91	12	Unknown_26	0.00	0.00	0.00	38
Corn_Unknown	1.00	1.00	1.00	71	Unknown_27	0.00	0.00	0.00	18
Fig_Blight	1.00	1.00	1.00	96	Unknown_28	0.27	0.18	0.22	76
Fig_Brown_Spot	1.00	0.98	0.99	53	Unknown_29	0.23	0.70	0.35	33
Fig_Rust	0.98	0.98	0.98	63	Unknown_3	0.00	0.00	0.00	55
Fig_Unknown	1.00	1.00	1.00	75	Unknown_30	0.25	0.48	0.33	66
Grape_Anthraco	0.95	0.94	0.94	124	Unknown_31	0.25	1.00	0.41	29
Grape_Black_Rot	0.75	0.98	0.85	124	Unknown_32	0.00	0.00	0.00	58
Grape_Blight_Isariopsis_Spot	0.74	1.00	0.85	107	Unknown_33	0.00	0.00	0.00	64
Grape_Brown_Spot	0.96	0.96	0.96	80	Unknown_34	0.50	0.02	0.04	47
Grape_Downy_Mildew	0.97	0.96	0.97	78	Unknown_35	0.00	0.00	0.00	200
Grape_Esca_Black_Measles	0.74	0.57	0.64	138	Unknown_36	0.29	0.80	0.43	15
Grape_Healthy	0.74	1.00	0.85	43	Unknown_37	0.25	0.04	0.06	56
Grape_Mites	1.00	1.00	1.00	63	Unknown_4	0.00	0.00	0.00	47

Grape_Powdery_Mildew	0.94	0.99	0.97	151	Unknown_5	0.31	0.12	0.18	41
Grape_Shot_Hole	0.99	1.00	1.00	106	Unknown_6	0.31	0.56	0.40	34
Grape_Unknown	1.00	0.98	0.99	193	Unknown_7	0.00	0.00	0.00	19
Orange_Haunglongbing_Greening	0.73	1.00	0.85	551	Unknown_8	0.00	0.00	0.00	43
Peach_Bacterial_Spot	0.73	1.00	0.85	230	Unknown_9	0.00	0.00	0.00	35
Peach_Healthy	0.77	1.00	0.87	36	Unknown_Lokat	0.98	1.00	0.99	63
Peach_Unknown	0.62	0.91	0.74	11	Unknown_Lokat_Spot	1.00	0.99	0.99	99
Pear_Black_Spot	0.99	1.00	1.00	113	Unknown_Persimmons_Brown_Spot	1.00	1.00	1.00	83
Pear_Fire_Blight	1.00	1.00	1.00	48	Walnut_Anthraco	1.00	1.00	1.00	81
Pear_Unknown	1.00	1.00	1.00	96	Walnut_Blotch	0.99	0.99	0.99	170
Pepper_Bacterial_Spot	0.77	0.99	0.87	100	Walnut_Gall_Mite	0.96	1.00	0.98	49
Pepper_Healthy	0.74	1.00	0.85	148	Walnut_Shot_Hole	0.99	0.98	0.99	114
Pepper_Spot	0.57	0.57	0.57	7	Walnut_Unknown	0.99	0.99	0.99	89
Pepper_Unknown	0.44	0.67	0.53	6	Wheat_Aphid	0.85	0.80	0.82	98
Potato_Early_Blight	0.71	0.87	0.78	112	Wheat_Black_Rust	0.69	0.80	0.74	64
Potato_Healthy	0.79	1.00	0.88	15	Wheat_Blast	0.82	0.88	0.85	72
Potato_Late_Blight	0.68	0.97	0.80	110	Wheat_Brown_Rust	0.85	0.86	0.85	134
Raspberry_Healthy	0.65	0.59	0.62	37	Wheat_Common_Root_Rot	0.79	0.80	0.79	69
Raspberry_Unknown	1.00	0.83	0.91	12	Wheat_Fusarium_Head_Blight	0.94	0.97	0.96	68
Soybean_Healthy	0.74	1.00	0.85	509	Wheat_Healthy	0.97	0.95	0.96	107
Soybean_Unknown	1.00	1.00	1.00	7	Wheat_Leaf_Blight	0.57	0.58	0.58	91
Squash_Powdery_Mildew	0.75	0.99	0.85	196	Wheat_Mildew	0.92	0.95	0.93	115
Strawberry_Healthy	0.71	1.00	0.83	45	Wheat_Mite	0.78	0.71	0.75	87
Strawberry_Scorch	0.74	0.97	0.84	111	Wheat_Septoria	0.93	0.97	0.95	121
Strawberry_Unknown	0.80	0.80	0.80	10	Wheat_Smut	0.90	0.90	0.90	138
Tomato_Bacterial_Spot	0.85	0.88	0.86	394	Wheat_Stem_Fly	0.91	0.97	0.94	30
Tomato_Curl	1.00	1.00	1.00	189	Wheat_Tan_Spot	0.62	0.58	0.60	84
Tomato_Early_Blight	0.94	0.70	0.80	301	Wheat_Yellow_Rust	0.98	0.99	0.99	137
Tomato_Fusarium_Wilt	0.98	1.00	0.99	41	macro avg	0.67	0.72	0.68	14260
					weighted avg	0.75	0.82	0.78	14260

Overall (weighted avg): Precision=0.75, Recall=0.82, F1=0.78, Accuracy=0.82 (n=14,260)

10. Model Artifacts Manifest

File	Description	Size
best_model.pth	Best checkpoint (epoch 5, full state dict + optimizer)	103 MB
final_model.pth	Last epoch checkpoint (epoch 12, weights only)	35 MB
model_scripted.pt	TorchScript traced model for deployment	36 MB
class_labels.json	Index → class name mapping (150 classes)	4 KB
training_config.json	Full hyperparameters + final metrics	1 KB
training_log.csv	Per-epoch metrics (12 rows)	1 KB
evaluation/classification_report.txt	Full sklearn classification report	13 KB
evaluation/evaluation_summary.json	Test metrics JSON	285 B
evaluation/training_curves.png	Loss, accuracy, F1, LR curves	331 KB
evaluation/confusion_matrix_top25.png	Top-25 confused class pairs	389 KB
evaluation/per_class_f1.png	Top/bottom 30 classes by F1	317 KB
evaluation/f1_histogram.png	F1 distribution histogram	63 KB
evaluation/confidence_distribution.png	Confidence + calibration plots	147 KB
splits/train.csv	Training split (114,076 rows)	38 MB
splits/val.csv	Validation split (14,260 rows)	4.8 MB
splits/test.csv	Test split (14,260 rows)	4.8 MB